



Trinity Grammar School

Mathematics Department

2014

HALF-YEARLY EXAMINATION

PRELIMINARY ASSESSMENT TASK 3

Year 11

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using **black** or **blue** pen
- Only approved calculators for this course are allowed in this task.
- Show all necessary working
- Write your **Name** and your **Class teacher** on the question paper and on any answer sheets or writing booklets used to write your responses to the questions submitted.
- If you do not attempt a question you must submit an answer sheet or writing booklet for that question clearly indicating **N/A** and your Name.
- **Preliminary Assessment Weighting: 30%**

(Year 12 only)

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Class Teacher:

.....

Name:

(Year 11 only)

.....

Do NOT write solutions on this question paper. Any working on this question paper will NOT be marked.

Date of examination:

Wednesday, 7 May 2014

Total marks – 70

Section I

10 marks

- Attempt all Questions
- Allow about **20** minutes for this section

Section II

60 marks

- Attempt all Questions
- Allow about **1** hours **40** minutes for this section

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Section I 10 marks

- Attempt Questions 1 – 10.
 - Allow about 20 minutes for this section.
 - Shade the correct response on the multiple-choice answer sheet for Questions 1 – 10.
 - Each question is worth 1 mark.
-

1 What is $\frac{1+\sqrt{3}}{5-2\sqrt{3}}$ as a fraction with a rational denominator?

(A) $\frac{-5-\sqrt{3}}{7}$

(B) $\frac{-5+\sqrt{3}}{13}$

(C) $\frac{11-7\sqrt{3}}{7}$

(D) $\frac{11+7\sqrt{3}}{13}$

2 Simplify $\frac{x^3-1}{x^2-1} \times \frac{x^2-4x-5}{4x^2+4x+4}$

(A) $\frac{(x-5)}{4}$

(B) $\frac{(x-1)}{4}$

(C) $\frac{(x+1)}{4}$

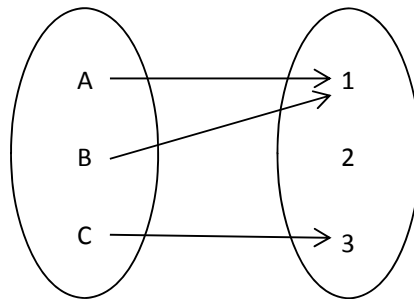
(D) $\frac{(x^2+x+1)}{4}$

3 What is the solution to the equation $81^{x+1} = \sqrt{3^x}$?

(A) $x = 1\frac{1}{7}$ (B) $x = -1\frac{1}{7}$ (A) $x = \frac{1}{4}$ (A) $x = -\frac{1}{4}$

4 Consider the relations below.

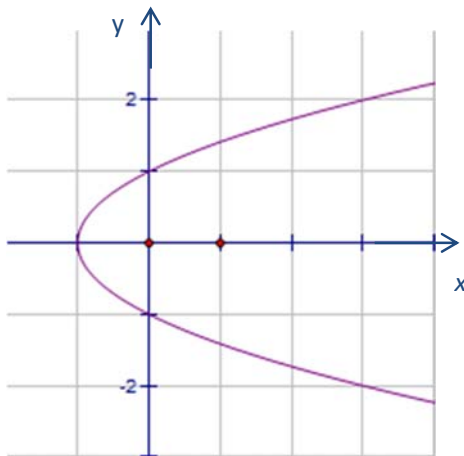
(i) Mapping:



(ii) Table:

Name	Ed	Carol	Peter	Don	Ed
Eye colour	Brown	Green	Blue	Brown	Grey

(iii) Graph:



Which of the following statements is true?

- (A) Only relation (i) is a function
- (B) Relation (i) and (iii) are functions
- (C) All three relations above are functions
- (D) None of the relations above are functions.

5 How many solutions are there to the equation $\sin \theta = \frac{1}{2}$ in the domain $0^\circ \leq \theta \leq 720^\circ$?

- (A) 1 solution
- (B) 2 solutions
- (C) 4 solutions
- (D) 8 solutions

6 What is the value of ab if the lines $ax + 2y = 6$ and $4y = bx - 9$ are perpendicular?

- | | |
|--------------------|-------------------|
| (A) $-\frac{1}{2}$ | (B) $\frac{1}{4}$ |
| (C) -8 | (D) 8 |

7 What is the solution to the inequality $\frac{3}{x-2} \leq 4$?

- (A) $x < -2$ and $x \geq -\frac{11}{4}$
- (B) $x > -2$ and $x \leq -\frac{11}{4}$
- (C) $x < 2$ and $x \geq \frac{11}{4}$
- (D) $x > 2$ and $x \leq \frac{11}{4}$

8 What is the acute angle between the lines $\sqrt{3}x - y = 1$ and $y = 0$?

- (A) 15°
- (B) 30°
- (C) 45°
- (D) 60°

- 9 What are the coordinates of the point that divides the interval joining the points $A(7,1)$ and $B(0,-6)$ **internally** in the ratio 4:3?

- (A) $(3,-3)$
- (B) $(3,-2)$
- (C) $(4,-2)$
- (D) $(4,-3)$

- 10 Point A is due south of a hill and the angle of elevation from A to the top of the hill is 35° . Another point B is on a bearing of 200° from the hill and the angle of elevation from B to the top of the hill is 46° . The distance AB is 220 m.

If the point O is at the base of the hill, and the hill has a height h metres, which of the following are the correct expressions for OA and OB ?

- (A) $OA = \frac{\tan 35^\circ}{h}$ and $OB = \frac{\tan 46^\circ}{h}$
- (B) $OA = \frac{h}{\tan 35^\circ}$ and $OB = \frac{h}{\tan 46^\circ}$
- (C) $OA = h \tan 35^\circ$ and $OB = h \tan 46^\circ$
- (D) $OA = 220 \tan 35^\circ$ and $OB = 220 \tan 46^\circ$

End of Section I
Section II commences on the next page

Section II 60 marks

- Attempt **Questions 11 – 14**
 - Allow about 1 hour and 40 minutes for this section
 - Answer each question in a SEPARATE writing booklet or answer sheet. Extra writing booklets are available.
 - In Questions 11 – 14, your responses should include all relevant mathematical reasoning and/or calculations.
-

Question 11 (15 marks)	Use a SEPARATE writing booklet	Marks
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(a) Write $(1 + \sqrt{5})^3$ in the form $a + b\sqrt{5}$, where a and b are integers.		2
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(b) Factorise $\frac{25}{x^2} - \frac{a^2}{b^2}$.		2
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(c) Solve $3(x-1) = \frac{4}{x+1} - 5$.		2
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(d) Increase the product of $(8x-5)$ and $(2x-7)$ by $(4x-1)$.		2
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(e) Simplify $\frac{4m^{-2}n}{(nm^2)^2} \times \frac{mn^{-1}}{(2mn)^3}$.		2
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(f) Simplify $\frac{5}{x^2-4} - \frac{3}{x-2} - \frac{2}{x+2}$.		2
--	--	----------

(g) Solve the simultaneous equations:		3
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$$a + b + c = 0$$

$$a - b + c = -4$$

$$2a - 3b - c = -1$$

Question 12 (15 marks)

Use a SEPARATE writing booklet

Marks

(a) If $f(x) = x^4 - 2x^2 - x$, **show** whether the function is odd, even or neither.

2

(b) Consider the piecemeal function $f(x) = \begin{cases} x+1, & \text{if } x < 0 \\ 1-x^2, & \text{if } 0 \leq x \leq 1. \\ \sqrt{x-1}, & \text{if } x \geq 1 \end{cases}$

3

- Find:
- (i) $f(-1)$
 - (ii) $f(2)$
 - (iii) Sketch $y = f(x)$ for $-1 \leq x \leq 2$.

(c) State the domain **and** range of the function $y = \sqrt{4+x^2}$.

2

(d) Find the equation of the line passing through the point $(-1,2)$ and perpendicular to the line $x - 2y + 2 = 0$.

3

(e) Graph the region bounded by the inequations $x + y \leq 0$ and $y \geq 2^x$.

2

(f) Show that the line $3x - 4y + 12 = 0$ does **not** touch or cut the circle $x^2 + y^2 = 1$.

3

Question 13 (15 marks)

Use a SEPARATE writing booklet

Marks

- (a) Simplify $\frac{\cos \theta}{1 + \cos \theta} + \frac{\sin^2 \theta}{(1 + \cos \theta)^2}$ as far as possible. 2
- (b) Find the exact value of $\sec(-60^\circ)\sin(120^\circ)\cos(240^\circ)$. 2
- (c) Solve $2\cos\theta + \sqrt{3} = 0$ for $0^\circ \leq \theta \leq 360^\circ$. 2
- (d) Prove $\cot\theta + \tan\theta = \operatorname{cosec}\theta \sec\theta$. 2
- (e) Find the area of a triangle with sides of 6 cm, 7 cm and 8 cm correct to one decimal place. 3
- (f) A ship leaves port *A* and sails 50 nautical miles due east to a port *B*.
It then proceeds a distance of 20 nautical miles at a bearing of 160° T to a port *C*.
- (i) Find the distance of *C* from *A* to the nearest nautical mile. 2
- (ii) Find the bearing of *C* from *A* to the nearest degree. 2

Question 14 (15 marks)

Use a SEPARATE writing booklet

Marks

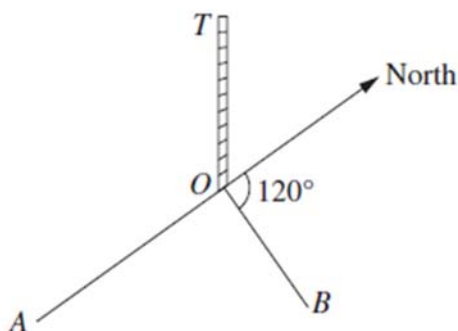
- (a) (i) Sketch the graph of $y = |2x - 1|$. **1**
(ii) Hence, or otherwise, solve $|2x - 1| \leq |x - 3|$. **3**

- (b) The acute angle between the lines $y = 3x + 5$ and $y = mx + 4$ is 45° . **2**
Find the **two** possible values of m .

- (c) Make x the subject of the formula $\frac{m}{n} = \frac{x - x_1}{x_2 - x}$. **3**

- (d) Let A be the point $(3, -1)$ and B be the point $(9, 2)$. Find the coordinates of the point P **2**
which divides the interval AB **externally** in the ratio $5:2$.

- (e) From a point A due south of a tower, the angle of elevation of the top of the tower T , is 23° .
From another point B , on a bearing of 120° from the tower, the angle of elevation of T is 32° .
The distance AB is 200 metres.



- (i) Copy or trace the diagram, adding the given information to your diagram. **1**
(ii) Show that the height (h) of the tower is given by **3**

$$h = \sqrt{\frac{40000}{\cot^2 23^\circ + \cot^2 32^\circ - \cot 23^\circ \cot 32^\circ}}.$$

End of Section II
End of Examination



Trinity Grammar School
Mathematics Department
2014

Year 11 Mathematics Extension 1
HALF-YEARLY EXAMINATION

Preliminary Assessment Task 3
Section I
Multiple Choice Answer Sheet

NSW Board of Studies Student Number

(Year 12 only)

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Name:

SOLUTIONS

(Year 11 only)

Class Teacher:

KESBY

- Write your answers for **Section I** on this answer sheet.
- **Shade completely only ONE answer for each question.**
- To change an answer, erase your previous mark completely and then record your new answer.

1	☐ (A)	☐ (B)	☐ (C)	☒ (D)
2	☒ (A)	☐ (B)	☐ (C)	☐ (D)
3	☐ (A)	☒ (B)	☐ (C)	☐ (D)
4	☒ (A)	☐ (B)	☐ (C)	☐ (D)
5	☐ (A)	☐ (B)	☒ (C)	☐ (D)
6	☐ (A)	☐ (B)	☐ (C)	☒ (D)
7	☐ (A)	☐ (B)	☒ (C)	☐ (D)
8	☐ (A)	☐ (B)	☐ (C)	☒ (D)
9	☒ (A)	☐ (B)	☐ (C)	☐ (D)
10	☐ (A)	☒ (B)	☐ (C)	☐ (D)

$$Q1. \quad \frac{1+\sqrt{3}}{5-2\sqrt{3}} = \frac{(1+\sqrt{3})}{5-2\sqrt{3}} \times \frac{(5+2\sqrt{3})}{5+2\sqrt{3}}$$

$$(D) \quad = \frac{5 + 2\sqrt{3} + 5\sqrt{3} + 2 \times 3}{25 - 12}$$

$$\begin{array}{|l} (2\sqrt{3})^2 = 4 \times 3 \\ = 12 \end{array}$$

$$= \frac{11 + 7\sqrt{3}}{13}$$

$$Q2. (A) \quad \frac{x^3-1}{x^2-1} \times \frac{x^2-4x-5}{4x^2+4x+4} = \frac{\cancel{(x-1)}(\cancel{x^2+x+1})}{\cancel{(x-1)}(x+1)} \times \frac{(x-5)\cancel{(x+1)}}{4(\cancel{x^2+x+1})}$$

$$= \frac{x-5}{4}$$

$$Q3. (B) \quad (81)^{x+1} = \sqrt{3^x}$$

$$(3^4)^{x+1} = (3^x)^{\frac{1}{2}}$$

$$3^{4x+4} = 3^{\frac{x}{2}}$$

$$4x+4 = \frac{x}{2}$$

$$8x+8 = x$$

$$7x+8 = 0$$

$$7x = -8$$

$$x = \frac{-8}{7}$$

$$x = -1\frac{1}{7}$$

Q4. (A) (i) Function ✓

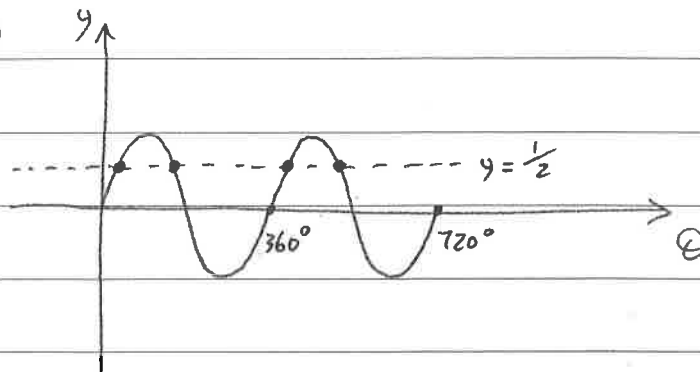
(ii) Not a function × * (Ed, Brown) & (Ed, Grey)

(iii) Not a function × * Does not satisfy the vertical line test.

Q5.

Consider $y = \sin \theta$

(c) 4 solutions



Q6. (D)

$$ax + 2y = 6$$

$$2y = -ax + 6$$

$$y = -\frac{a}{2}x + 3$$

$$m_1 = -\frac{a}{2}$$

$$4y = bx - 9$$

$$y = \frac{b}{4}x - \frac{9}{4}$$

$$m_2 = \frac{b}{4}$$

For perpendicular lines

$$m_1 \times m_2 = -1$$

$$-\frac{a}{2} \times \frac{b}{4} = -1$$

$$-\frac{ab}{8} = -1$$

$$\frac{ab}{8} = 1$$

$$ab = 8.$$

Q7. (c)

$$(x-2)^2$$

$$\frac{3}{(x-2)^2}$$

$$\leq 4(x-2)^2$$

$$(x \neq 2)$$

$$3(x-2) \leq 4(x-2)^2$$

$$0 \leq 4(x-2)^2 - 3(x-2)$$

$$4(x-2)^2 - 3(x-2) \geq 0$$

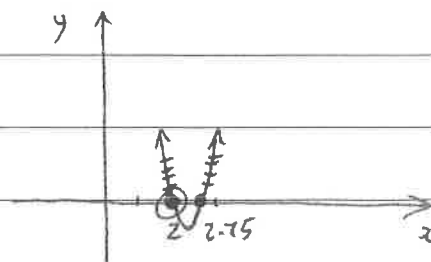
$$(x-2)(4(x-2) - 3) \geq 0$$

$$(x-2)(4x-11) \geq 0$$

Solution is

$$\text{and } x \geq \frac{11}{4}$$

$$x < 2$$



Do NOT write anything, or make any marks below this line or outside borders

Q8. (D)

$$\sqrt{3}x - y = 1$$

$$\sqrt{3}x = y + 1$$

$$\sqrt{3}x - 1 = y$$

$$y = \sqrt{3}x - 1$$

$$m_1 = \sqrt{3}$$

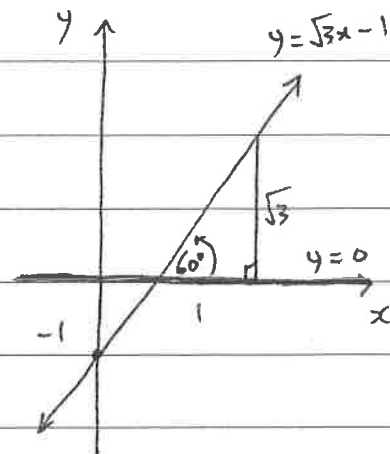
$$m_1 = \tan \theta$$

$$\sqrt{3} = \tan \theta$$

$$\tan \theta = \sqrt{3}$$

$$\theta = \tan^{-1}(\sqrt{3})$$

$$\theta = 60^\circ$$



E^{ak}

Q9. (A)

$$m:n = 4:3$$

$$\frac{mx_2 + nx_1}{m+n}$$

$$x = \frac{mx_2 + nx_1}{m+n}$$

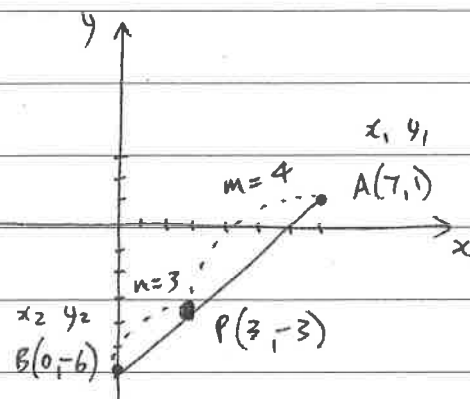
$$\frac{4(0) + 3(7)}{4+3}$$

$$x = \frac{4+3}{0+21}$$

$$x = \frac{7}{21}$$

$$x = \frac{1}{3}$$

$$x = \frac{1}{3}$$



$$\frac{my_2 + ny_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$\frac{4(-6) + 3(1)}{4+3}$$

$$y = \frac{-24+3}{7}$$

$$y = \frac{-21}{7}$$

$$y = -3$$

$$y = -3$$

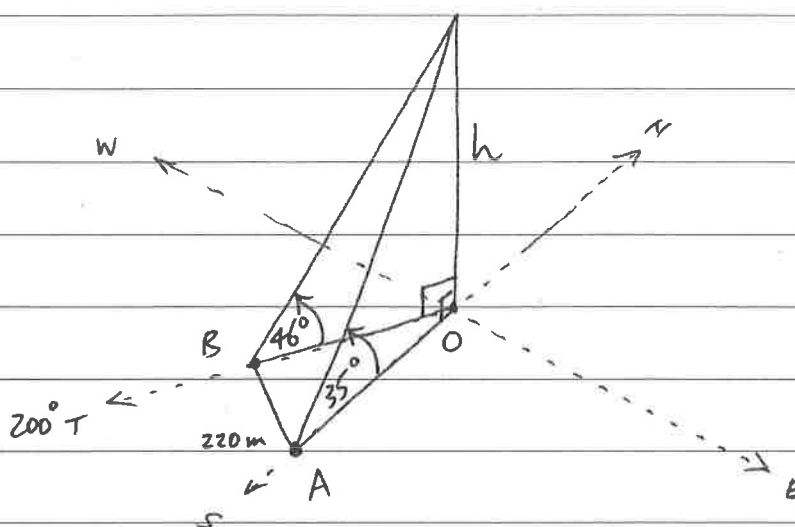
$$\therefore P\left(\frac{1}{3}, -3\right)$$



Tick this box if you have continued this answer in another booklet.

Do NOT write anything, or make any marks below this line or outside borders

E* Q10. (B)



$$\tan 35^\circ = \frac{h}{OA} \quad \tan 46^\circ = \frac{h}{OB}$$

$$OA \times \tan 35^\circ = h \quad OB \times \tan 46^\circ = h$$

$$OA = \frac{h}{\tan 35^\circ} \quad OB = \frac{h}{\tan 46^\circ}$$

$$\begin{aligned}
 \textcircled{11} \quad (a) \quad (1 + \sqrt{5})^3 &= (1 + \sqrt{5})^2 \times (1 + \sqrt{5}) \\
 &= (1 + 2\sqrt{5} + 5)(1 + \sqrt{5}) \\
 &= (2\sqrt{5} + 6)(1 + \sqrt{5}) \quad \textcircled{1} \\
 &= 2\sqrt{5} + 2\sqrt{5} \times \sqrt{5} + 6 + 6\sqrt{5} \\
 &= 2\sqrt{5} + 10 + 6 + 6\sqrt{5} \\
 &= 16 + 8\sqrt{5}. \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \frac{25}{x^2} - \frac{a^2}{b^2} &= \left(\frac{5}{x}\right)^2 - \left(\frac{a}{b}\right)^2 \quad \textcircled{1} \\
 &= \left(\frac{5}{x} + \frac{a}{b}\right)\left(\frac{5}{x} - \frac{a}{b}\right). \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad 3(x-1) &= \frac{4}{(x+1)} - 5 \\
 3(x-1)(x+1) &= 4 - 5(x+1) \\
 3(x^2-1) &= 4 - 5x - 5 \\
 3x^2 - 3 &= -5x - 1 \\
 3x^2 + 5x - 2 &= 0 \quad \textcircled{1} \\
 (3x-1)(x+2) &= 0 \\
 \therefore x &= \frac{1}{3}, -2. \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad (8x-5)(2x-7) + (4x-1) &= 16x^2 - 56x - 10x + 35 + 4x - 1 \quad \textcircled{1} \\
 &= 16x^2 - 62x + 34. \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad \frac{4m^{-2}n}{(nm^2)^2} \times \frac{mn^{-1}}{(2mn)^3} &= \frac{4 \times m^{-2} \times m^1 \times n^1 \times n^{-1}}{n^2 \times m^4 \times 8 \times m^3 \times n^3} \\
 &= \frac{4 \times m^{-1}}{8 \times m^7 \times n^5} \quad \textcircled{1} \\
 &= \frac{1}{2} \times \frac{1}{m^8} \times \frac{1}{n^5} = \frac{1}{2m^8n^5}. \quad \textcircled{1}
 \end{aligned}$$

Do NOT write anything, or make any marks below this line or outside borders

$$\begin{aligned}
 \textcircled{11} \quad (f) \quad \frac{5}{x^2-4} - \frac{3}{x-2} - \frac{2}{x+2} &= \frac{5 - 3(x+2) - 2(x-2)}{(x+2)(x-2)} \\
 &= \frac{5 - 3x - 6 - 2x + 4}{(x+2)(x-2)} \\
 &= \frac{-5x + 3}{(x+2)(x-2)}.
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{11} \quad (g) \quad a + b + c &= 0 & \text{---} \textcircled{1} \\
 a - b + c &= -4 & \text{---} \textcircled{2} \\
 2a - 3b - c &= -1 & \text{---} \textcircled{3}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{1} - \textcircled{2}: \quad b - (-b) &= 0 - (-4) \\
 2b &= 4 \\
 \underline{b} &= 2 & \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} + \textcircled{3} \quad 3a - 4b &= -5 \\
 \text{Subst. } b=2 \quad 3a - 4(2) &= -5 \\
 3a - 8 &= -5 \\
 3a &= 3 \\
 \underline{a} &= 1 & \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Subst. } a=1 \text{ \& } b=2 \text{ into } \textcircled{1}: \quad 1 + 2 + c &= 0 \\
 3 + c &= 0 \\
 \underline{c} &= -3 & \textcircled{1}
 \end{aligned}$$

$$\therefore a=1, b=2 \text{ \& } c=-3.$$

(12)

(a)

$$f(-x) = (-x)^4 - 2(-x)^2 - (-x)$$

$$= x^4 - 2x^2 + x$$

$$\neq x^4 - 2x^2 - x \quad (1)$$

$$\neq -(x^4 - 2x^2 - x)$$

$$\text{ie. } f(-x) \neq f(x) \text{ and } f(-x) \neq -f(x)$$

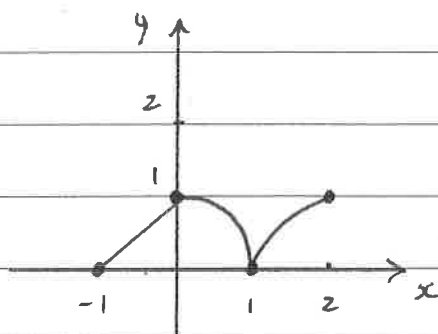
$\therefore$ function is neither even nor odd. (1)

(b)

$$f(-1) = -1 + 1 = 0$$

$$f(2) = \sqrt{2-1} = \sqrt{1} = 1$$

} (1)

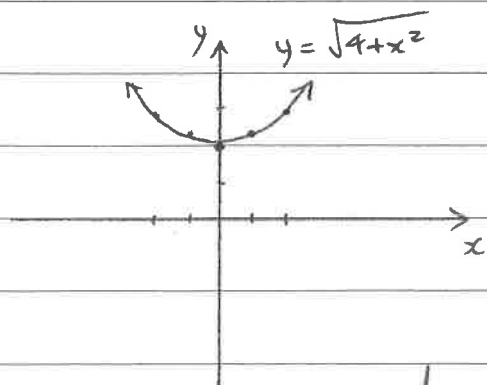


(2)

(c) y-intercept: $x=0$, $y = \sqrt{4+(0)^2}$

$$y = \sqrt{4}$$

$$y = 2 \quad (0, 2)$$



x	0	1	2
y	2	$\sqrt{5}$	$\sqrt{8}$

domain: all real x (1)

range: $y \geq 2$ (1)

(d)

$$x - 2y + 2 = 0$$

$$x + 2 = 2y$$

$$2y = x + 2$$

$$y = \frac{1}{2}x + 1$$

$$m_1 = \frac{1}{2}$$

$$m_2 = -2$$

 x_1, y_1

$$(-1, 2) \quad m_2 = -2$$

$$y - y_1 = m_2(x - x_1)$$

$$y - 2 = -2(x - (-1)) \quad (1)$$

$$y - 2 = -2(x + 1)$$

$$y - 2 = -2x - 2$$

$$y = -2x \quad (1)$$

(e)

Consider $x + y = 0$

$$y = -x$$

$$y = 2^x$$

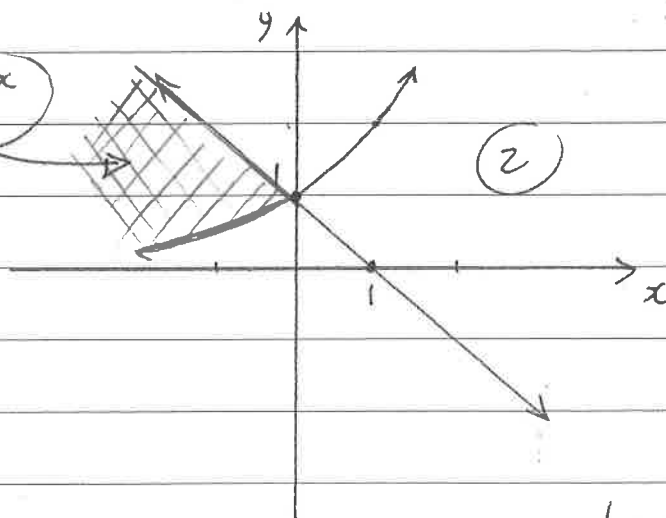
Test $(0, 0)$: $y \geq 2^x$

$$0 \geq 2^0$$

$$0 \geq 1$$

False.

$$x + y \leq 0 \wedge y \geq 2^x$$



(2)

Test $(0, 0)$: $x + y \leq 0$

$$0 + 0 \leq 0$$

True.

(f)

$$3x - 4y + 12 = 0$$

$$a = 3$$

$$b = -4$$

$$c = 12$$

$$x_1, y_1$$

$$(0, 0)$$

$$x^2 + y^2 = 1$$

Circle, centre $(0, 0)$

radius = 1 unit

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$d = \frac{|3(0) + (-4)(0) + 12|}{\sqrt{(3)^2 + (-4)^2}} \quad (1)$$

$$d = \frac{12}{\sqrt{25}}$$

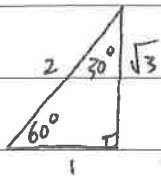
$$d = \frac{12}{5} = 2.4 \quad (1)$$

Now, $d > \text{radius} \therefore$ line does not touch or cut circle. (1)

Do NOT write anything, or make any marks below this line or outside borders

13 (a)
$$\frac{\cos \theta}{1 + \cos \theta} + \frac{\sin^2 \theta}{(1 + \cos \theta)^2} = \frac{\cos \theta (1 + \cos \theta) + \sin^2 \theta}{(1 + \cos \theta)^2}$$
$$= \frac{\cos \theta + \cos^2 \theta + \sin^2 \theta}{(1 + \cos \theta)^2}$$
$$= \frac{\cos \theta + 1}{(\cos \theta + 1)^2}$$
$$= \frac{1}{\cos \theta + 1} \quad (1)$$

(b)
$$\sec(-60^\circ) = \frac{1}{\cos(-60^\circ)} = \frac{1}{\cos 300^\circ} = \frac{1}{\cos 60^\circ} = \frac{1}{(\frac{1}{2})} = 2.$$
$$\sin(120^\circ) = \sin 60^\circ = \frac{\sqrt{3}}{2}.$$
$$\cos(240^\circ) = -\cos 60^\circ = -\frac{1}{2}.$$

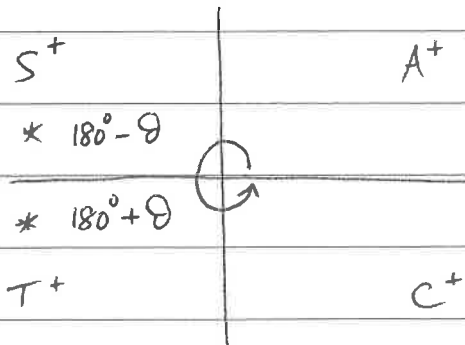


$\cos 60^\circ = \frac{1}{2}$

$$\therefore \sec(-60^\circ) \times \sin(120^\circ) \times \cos(240^\circ) = 2 \times \frac{\sqrt{3}}{2} \times -\frac{1}{2}$$
$$= -\frac{\sqrt{3}}{2} \quad (1)$$

(c)
$$2 \cos \theta = -\sqrt{3}$$
$$\cos \theta = -\frac{\sqrt{3}}{2}$$
$$\theta = 180^\circ - 30^\circ, 180^\circ + 30^\circ$$
$$\theta = 150^\circ, 210^\circ$$

(1) (1)

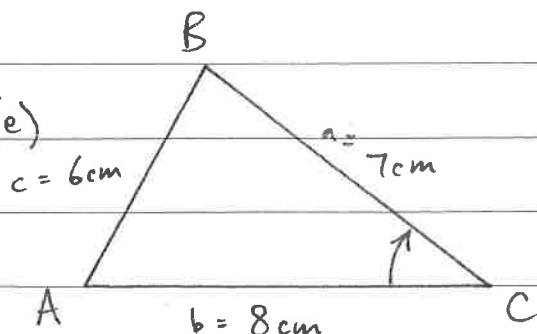


(base angle = 30°)

(d) L.H.S.
$$= \cot \theta + \tan \theta$$
$$= \frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta}$$
$$= \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cdot \cos \theta}$$
$$= \frac{1}{\sin \theta \cos \theta}$$
$$= \frac{1}{\sin \theta} \times \frac{1}{\cos \theta} = \operatorname{cosec} \theta \cdot \sec \theta = \text{R.H.S.}$$

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(13) (e)



$$A = \frac{1}{2} ab \sin C$$

$$A = \frac{1}{2} \times 7 \times 8 \times \sin(46.56746344^\circ)$$

$$A = 20.33316257...$$

$$\text{Area} = 20.3 \text{ cm}^2 \text{ (1d.p.)}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

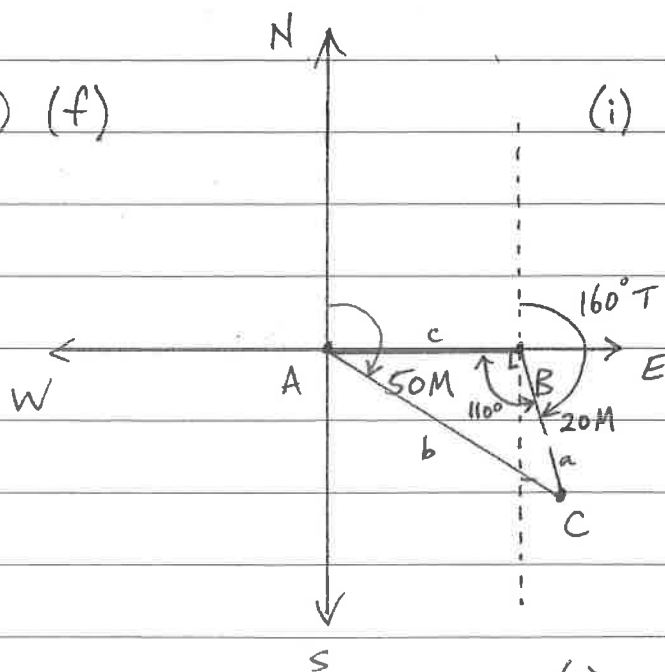
$$\cos C = \frac{7^2 + 8^2 - 6^2}{2(7)(8)}$$

$$\cos C = 0.6875$$

$$\angle C = \cos^{-1}(0.6875)$$

$$\angle C = 46.56746344^\circ$$

(13) (f)



$$(i) b^2 = a^2 + c^2 - 2ac \cos B$$

$$b^2 = 20^2 + 50^2 - 2(20)(50) \cos 110^\circ$$

$$b^2 = 3584.040287...$$

$$b = 59.86685466...$$

$$\text{AC} = 60 \text{ nautical miles}$$

$$(ii) \frac{\sin A}{20} = \frac{\sin 110^\circ}{60}$$

$$\sin A = \frac{20 \sin 110^\circ}{60}$$

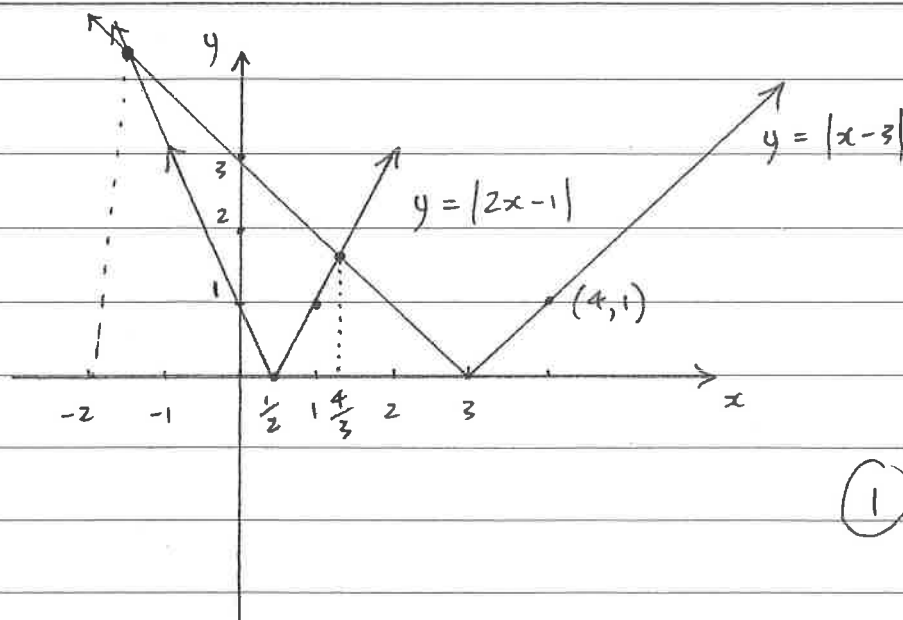
$$\sin A = 0.3132...$$

$$\angle A = 18^\circ \text{ (nearest degree)}$$

$$\therefore \text{bearing of C from A is } 108^\circ \text{ T.}$$

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14 (a) (i)



①

(a) (ii) Consider $|2x-1| = |x-3|$

$$2x-1 = x-3 \quad \text{OR} \quad 2x-1 = -(x-3)$$

$$\underline{x = -2} \quad \text{①}$$

$$2x-1 = -x+3$$

$$3x = 4$$

$$\underline{x = \frac{4}{3}} \quad \text{①}$$

$\therefore$ from the graph,

$$|2x-1| \leq |x-3| \quad \text{for} \quad \boxed{-2 \leq x \leq \frac{4}{3}} \quad \text{①}$$

14

(b)

$$y = 3x + 5$$

$$y = mx + 4$$

$$\theta = 45^\circ$$

$$m_1 = 3$$

$$m_2 = m$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{3 - m}{1 + 3m} \right|$$

$$\text{Now, } 1 = \frac{3-m}{1+3m} \quad \text{or} \quad -1 = \frac{3-m}{1+3m}$$

$$1+3m = 3-m$$

$$-(1+3m) = 3-m$$

$$4m = 2$$

$$-1-3m = 3-m$$

$$\underline{m = \frac{1}{2}} \quad (1)$$

$$-2m = 4$$

$$\underline{m = -2} \quad (1)$$

$$(14) \quad (c) \quad \frac{m}{n} = \frac{x-x_1}{x_2-x_1}$$

$$m(x_2-x) = n(x-x_1) \quad (1)$$

$$mx_2 - mx = nx - nx_1$$

$$mx_2 + nx_1 = mx + nx \quad (1)$$

$$mx_2 + nx_1 = x(m+n)$$

$$\frac{mx_2 + nx_1}{m+n} = x$$

$$\therefore x = \frac{mx_2 + nx_1}{m+n} \quad (1)$$

$$(14) \quad (d) \quad \begin{matrix} x_1 & y_1 & & x_2 & y_2 \\ A(3, -1) & & B(9, 2) & & m:n = 5:-2 \end{matrix}$$

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{5(9) + (-2)(3)}{5 + (-2)}$$

$$y = \frac{5(2) + (-2)(-1)}{5 + (-2)}$$

$$x = \frac{45-6}{3}$$

$$y = \frac{10+2}{3}$$

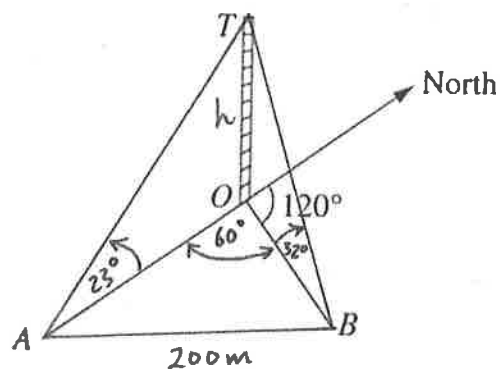
$$x = 13$$

$$y = 4$$

$$\therefore P(13, 4) \quad (1) \quad (1)$$

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14 (e) (i)



(e) (ii) Let $TO = h$ metres

$$\text{In } \triangle AOT, \tan 23^\circ = \frac{h}{AO}$$

$$\therefore AO = \frac{h}{\tan 23^\circ}$$

$$\text{In } \triangle BOT, \tan 32^\circ = \frac{h}{BO}$$

$$\therefore BO = \frac{h}{\tan 32^\circ}$$

Note that $\angle AOB = 60^\circ$.

Using the cosine rule,

$$AB^2 = AO^2 + BO^2 - 2(AO)(BO)\cos 60^\circ$$

$$200^2 = \left(\frac{h}{\tan 23^\circ}\right)^2 + \left(\frac{h}{\tan 32^\circ}\right)^2 - 2\left(\frac{h}{\tan 23^\circ}\right)\left(\frac{h}{\tan 32^\circ}\right)\cos 60^\circ$$

$$\therefore 40\,000 = \frac{h^2}{\tan^2 23^\circ} + \frac{h^2}{\tan^2 32^\circ} - \frac{h}{\tan 23^\circ} \times \frac{h}{\tan 32^\circ}$$

$$\therefore 40\,000 = h^2 \left[\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 32^\circ} - \frac{1}{\tan 23^\circ \times \tan 32^\circ} \right]$$

$$\text{So, } h^2 = \frac{40\,000}{\left[\frac{1}{\tan^2 23^\circ} + \frac{1}{\tan^2 32^\circ} - \frac{1}{\tan 23^\circ \times \tan 32^\circ} \right]}$$

$$\therefore h = \sqrt{\frac{40\,000}{\cot^2 23^\circ + \cot^2 32^\circ - \cot 23^\circ \cot 32^\circ}}$$



Tick this box if you have continued this answer in another booklet.

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